

ANSWERS

1.

Question	Answer	Marks
3(a)(i)	it / lamina is in the gravitational field (of Earth)	B1
3(a)(ii)	$(F =) mg$ or 0.050×9.8 or 50×9.8 or 490	C1
	0.49 (N)	A1
3(b)(i)	$(M =) F \times x_{\perp}$ or 0.49×8.0 or 0.49×0.080	C1
	0.039 (N m)	A1
3(b)(ii)	it / lamina moves clockwise (about the pivot)	B1
	it speeds up and then slows down or it oscillates (with decreasing amplitude) or overshoots	B1
	(stops with) G vertically below H	B1

2.

Question	Answer	Marks
1(a)(i)	longer	B1
	thinner or smaller (cross-sectional) area	B1
1(a)(ii)	elastic potential or strain (energy)	B1
1(b)(i)	sum of / total / net / resultant moment is zero	M1
	when (the object is) in equilibrium	A1
1(b)(ii)	distance of rubber band to pivot is smaller than the distance of the mass (to pivot)	B1
1(b)(iii)	$4 \times 18 = F \times 2$	C1
	36 N	A1
1(b)(iv)	greater and the weight (of XY) contributes to the (clockwise) moment/turning effect or the moment of the rubber band force is larger	B1

3.

Question	Answer	Marks
1(a)	$(p =) F / A$ or $2000 / (2 \times 0.040)$ or 50 000 (P–a)	C1
	25 000 Pa	A1
1(b)(i)	symmetrical left-to-right or symmetrical front-to-back or bench is stable	B1
	because centre of mass is in middle (horizontally) or centre of mass is within the base	B1
1(b)(ii)	(moment =) Fx or Wy or $2000 \times (1.1 - 0.25)$ or 2000×0.85 clockwise moment = anticlockwise moment or $Fx = Wy$ or	C1
	$Fx = 2000 \times (1.1 - 0.25)$ or $Fx = 2000 \times 0.85$ or $(F =) 2000 \times 0.85 / 0.25$	C1
	6800 N	A1

4.

Question	Answer	Marks
8(a)	$F \times \text{distance / displacement}$	M1
	perpendicular distance to line of action of force shown on Fig. 7.1	A1
8(b)(i)	for an object in equilibrium	B1
	(sum of) clockwise moments is equal to (sum of) anticlockwise moments or no <u>resultant</u> moment	B1
8(b)(ii)	equipment shown in diagram (e.g. rule, two loads, pivot)	B1
	<u>what is done</u> to achieve <u>balance</u> (e.g. balance rule on pivot, one load each side of pivot)	B1
	how principle is verified (e.g. calculate two opposite moments and find they are equal)	B1
8(c)(i)	80 N	B1
8(c)(ii)	(point) where the (whole) weight of an object acts / seems to act or point where the object balances	B1

Question	Answer	Marks
8(c)(iii)	mass not uniformly distributed along length or more mass at the one end	B1
8(c)(iv)	$(\Gamma =) 80 \times 0.90$	C1
	72 N m	A1
8(c)(v)	moment / 0.54 or 209 (N) or 41.1 / 0.54	C1
	76 N	A1
	downwards	B1

5.

Question	Answer	Marks
3(a)	(mass =) meter reading / g meter reading / gravitational field (strength) meter reading / acceleration due to gravity	B1
3(b)(i)	(sum of) clockwise moments = (sum of) anticlockwise moments (about any point)	B1
Question	Answer	Marks
3(b)(ii)	any attempt at using moments, e.g. $3 \times 30 = F \times 20$	C1
	4.5 N	A1
3(c)	force upwards (on brick due to water) or force due to water (on brick) or force downwards (on brick) is reduced / counteracted (by water) or force exerted <u>by</u> brick (on rule) reduced or force / tension in string reduced	B1
	(force caused by) <u>pressure</u> (of water) or reduced <u>anticlockwise moment</u>	B1

6.

Question	Answer	Marks
3(a)(i)	$(\Gamma =) Fx_{\perp}$ or 25×0.72	C1
	18 N m	A1
3(a)(ii)	$(WD =) Fx_{\parallel}$ or $F\pi r / 2$ or $25 \times 4.5 / 4$ or 25×1.13	B1
	28 J	B1
3(b)	moment due to force P is greater	B1
	distance of X to hinge is greater than distance from where force F acts	B1

7.

Question	Answer	Marks
1(a)	$(m =) \rho V$ or 1.8×200 or 360	C1
	$(1.8 \times 200) + 50$ or $360 + 50$ or 360 or candidate's mass + 50	C1
	410 g or 0.41 kg	A1
1(b)(i)	(place / region) where the mass appears to be concentrated or where the object balances	B1
1(b)(ii)	88 (cm) seen or $(84 + 92) / 2$ seen or 30 (cm) seen or 8(.0 cm) seen or 164 (g)	C1
	$m_1 g x_1 = m_2 g x_2$ or $F_1 x_1 = F_2 x_2$ or $W_1 x_1 = W_2 x_2$ or $(m_1 =) m_2 x_2 / x_1$ or $410 \times (88 - 80) / 30$ or 164 (g)	C1
	110 g or 0.11 kg	A1

8.

Question	Answer	Marks
2(a)	Hang card from pin / nail through hole and wait until stationary	B1
	using plumb line from hole mark line (on card)	B1
	repeat with another hole	B1
2(b)(i)	centre of mass / C falls outside base	B1
	weight creates moment / turning effect	B1
2(b)(ii)	lower centre of gravity or wider / longer base	C1

9.

- (a) substance or matter B1
rest or motion B1
- (b)(i) (a region of space) where a mass experiences a force (due to gravitational attraction) B1
- (b)(ii) $(m =) \rho V$ or $1000 \times 2.4 \times 10^{-2}$ or 24 (kg) C1
or C1
250 N
25 (kg) 240 (N) A1
- (c)(i) $(\Gamma =) Fx$ or 250×0.12 C1
30 N m A1
- (c)(ii) 75 N B1

10.

- (a) force $\times$ distance C1
ignore force into distance
- force $\times$ perpendicular distance (from line of action to point / pivot) A1
- (b) any moment calculation seen, e.g. $F \times 22 = 80 \times 4$ C1
15 N A1

11.		
(a) (i)	20 N	B1
(ii) 1.	$(\Gamma =)Fd$ or 20×0.35 or 20×0.70 or 14	C1
	7.0 N m	A1
2.	friction (at hinge/seal) or air resistance or to cause an initial acceleration	B1
(b)	(for other directions) perpendicular distance is less	B1
12.		
(a)	260 N	B1
(b) (i)	for a body in equilibrium	B1
	(total) clockwise moment = (total) anticlockwise moment	B1
(ii)	$F_1d_1 = F_2d_2$ or 260×0.35 or 91 or $F \times 0.65$	C1
	$260 \times 0.35 = F \times 0.65$ or $260 \times 0.35/0.65$ or $91 = F \times 0.65$ or $91/0.65$	C1
	140 N	A1
13.		
(a)	(point) C immediately above tip of pivot (and in middle(vertically) of screwdriver (± 1 mm))	B1
(b) (i)	0.64 N	B1
(ii)	arrow W vertically downwards through candidate's C or pivot	B1
(c)	no resultant force or upward force = downward force or force left = force right	B1
	no resultant moment (or force) or clockwise moment = anticlockwise moment	B1
14.		
(a) (i)	force moves through a distance (in same direction)	B1
(ii)	chemical (potential) energy	B1
(b) (i)	480 Nm	B1
(ii)	attempt to apply moments with two forces and distances	C1
	400 N	A1
15.		
(a) (i)	Fd or 2.5×0.18	C1
	0.45 N m	A1
(ii)	force not applied at right angles to the tap	B1
(b)	long(er) distance needs small(er) force (for same moment) or inversely related/proportional	B1

- 16.
- (a) force $\times$ distance **or** $F \times d$ with F and d defined **or** $F \times d_{\text{perp}}$
force $\times$ perpendicular distance **or** $F \times d_{\text{perp}}$ with F and d_{perp} defined C1
A1
- (b) (i) 1. $6 \times 750 \times 1.2$ **or** 750×1.2 **or** 900 C1
5400 Nm A1
2. mgh **or** $350 \times 10 \times 160$ **or** $350 \times 10 \times 1.6$ C1
 $350 \times 10 \times 1.6$ **or** 5.6×10^5 C1
5600J A1
- (ii) friction at axle/boat **or** drag due to water
or chain lifted also B1
heat produced (**ignore** in sailors) **or** work done against friction/drag
or work done raising chain B1
- (iii) same amount of work done **or** $P = E/t$ **or** $P = WD/t$ B1
in less time **or** power inversely proportional to time (**ignore** faster rate) B1
- (c) clear/labelled diagram with ruler, fulcrum, at least two weights B1
any three of the following points made in words:
balance ruler (on its own)
place weights on ruler so it balances
clockwise and anticlockwise moments equal **or** net moment zero
repeat (apply similar principles to other possible methods) B3
- 17.
- (a) $F_1x_1 = F_2x_2$ **or** $550 \times (0.86 \text{ or } 86)/(1.1 \text{ or } 110)$ C1
430 N A1
- (b) **both** moments increase C1
girl's moment increases more **or** girl's moment $>$ brother's
or anticlockwise moment greater A1
see-saw tips down on girl's side B1
- 18.
- (a) force $\times$ distance M1
perpendicular distance A1
- (b) (i) $T \times 8$ or 2000×2 seen C1
500 N A1
- (ii) (two forces) equal (in magnitude) B1
(two forces) opposite (in direction) B1
- 19.
- (a) total/resultant **moment** zero **or** (sum of) clockwise = anticlockwise **moment** B1
- (b) F_1d_1 **or** F_2d_2 seen in any form C1
8(.0) N A1
- (c) $4 + 1.2$ or 5.2 seen C1
2.8 N ecf (b) i.e. accept $5.2 - (b)$ or $(b) - 5.2$ A1

20.		
(a)	appropriate apparatus e.g. ruler, weights, fulcrum	B1
	action e.g. balance weights on each side	B1
	one of: force/mass $\times$ distance or calculate moment	B1
	vary or repeat	B1
(b)	$F \times d$ or 8.0×0.15	C1
	1.2 Nm (not J)	A1
21.		
(a)	(i) force $\times$ distance	C1
	force $\times$ perpendicular distance (from the axis)	A1
	(ii) 8200×0.05	C1
	410 Nm	A1
	(iii) (perpendicular) distance reduced/force not perpendicular/only a component of the force is perpendicular	B1
22.		
(a)	(i) ($M =$) force $\times$ perpendicular distance or 840×5	
	(formula mark can be scored if not given in 3(a)(ii))	C1
	4200 Nm	A1
	(ii) 350 N or (a)(i)/12 and calculated	B1
	(iii) weight of ladder/hose or friction at P/pivot/axle	
	(not air resistance; ign. friction)	B1
23.		
(a)	1.(0) m	B1
(b)	(i) (for an object in) equilibrium/balance	B1
	$W_1x = W_2y$ (clear) or anticlockwise moment/torque/turning force = clockwise moment/torque/turning force	B1
	(ii) $18\,000 \times 1.0 = T \times 0.5$	C1
	36 000 N	A1